

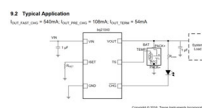
Calcs:

<https://docs.google.com/spreadsheets/d/1r5jx09jgPo7Qa4dIGZ32ZsXNAvsbSHH8eOILfTHGSew/edit?usp=sharing>

$RISET = [540\Omega / 0.45A] = 1.2\text{ k}\Omega$

To Disable the temp sense function, use a fixed 10k resistor between the TS (terminal 1) and Vss.

LED Status: connect a 1.5k Ω resistor in series with a LED between the OUT terminal and the CHG terminal.



Battery Life

Battery: 500mAh, 3.7V

Discharge

Running

500mAh/20mA = 25h

$500mAh / (30mA \cdot 6.4V / 3.7V / 0.8 + 1.65mA + 2.1mA) = 7.2h$

$500mAh \cdot 6.4V / 0.8 / 3.7V = 65mA + 1.65mA + 2.1mA = 68.8mA$

20mA = 47mA = 10.6h

LED driver quiescent current = 1.65mA max

LED driver efficiency (5% duty) = 80%

MCU quiescent current: 2.1mA

Shutdown

$500mAh / (.0071) = 70.422h = 8y$

MCU shutdown current: 6.1uA

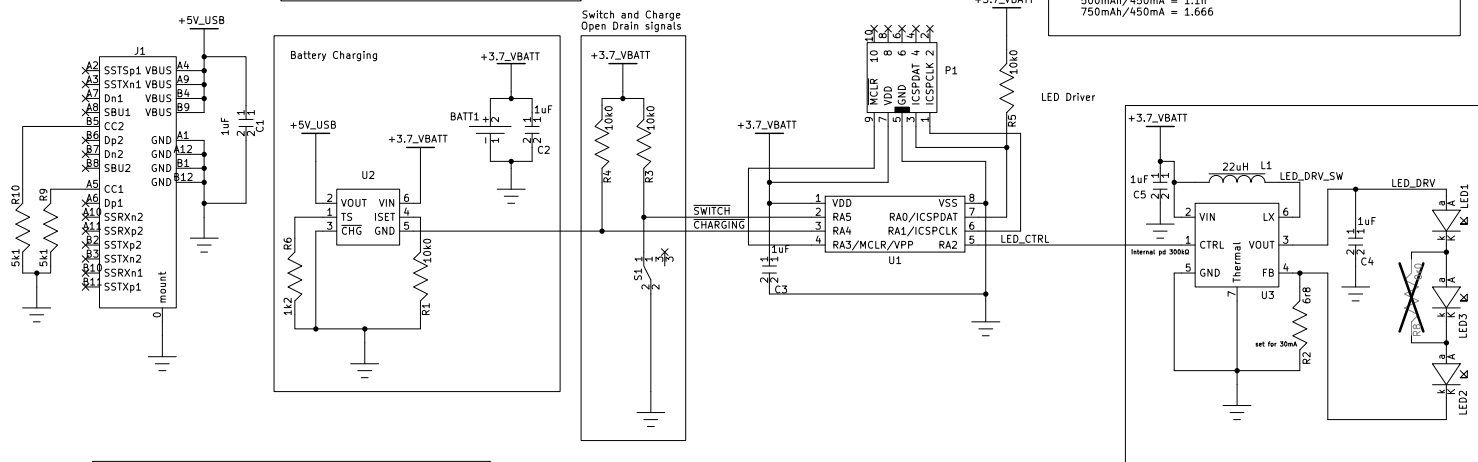
LED Driver Shutdown current: 1uA

cap leakage, etc.: ?

Charging

$500mAh / 450mA = 1.1h$

$750mAh / 450mA = 1.66h$



PENDING CHANGES Rev 0.2

[X] - Measure Battery Voltage? - move measurement to different pin

[X] - Add another LED

[X] - Battery Connector

[X] - USBC - 5k1 R from CC to GND

[X] - 0402 components for more space, smaller MCU

Test Points

